Appendices: Theoretical and Empirical Analysis of TSVD-Based Word Embedding Compression

A Theoretical Analysis of TSVD-Induced Perturbation

To further explain why the truncated singular value decomposition (TSVD) adopted in this paper can effectively reduce the number of parameters in the word embedding layer while avoiding excessive performance degradation, this appendix provides a perturbation propagation analysis from matrix reconstruction error to token-level representation error and then to downstream output error. The purpose is to show that the perturbation introduced by TSVD into the word embedding layer is controlled and quantifiable.

A.1 Notation and Problem Setup

Let the BERT word embedding matrix be

$$W_t \in \mathbb{R}^{|V| \times H}, \tag{A.1}$$

where $|V|$ denotes the vocabulary size and H denotes the hidden dimension. Let $r = \mathrm{rank}(W_t)$. Using the compact singular value decomposition, we write

$$W_t = U_r \Sigma_r V_r^\top, \tag{A.2}$$

where $U_r \in \mathbb{R}^{|V| \times r}$ and $V_r \in \mathbb{R}^{H \times r}$ have orthonormal columns, and $\Sigma_r = \mathrm{diag}(\sigma_1, \ldots, \sigma_r)$ contains singular values satisfying $\sigma_1 \geq \cdots \geq \sigma_r > 0$. For a truncation rank $P \leq r$, retaining the top P singular values and their corresponding singular vectors gives the rank-P TSVD approximation

$$\hat{W}_t = U_P \Sigma_P V_P^\top. \tag{A.3}$$

The original word embedding matrix can therefore be approximately factorized into the product of two low-rank matrices:

$$W_t \approx W_{t_1} W_{t_2} = (U_P \Sigma_P) V_P^\top, \tag{A.4}$$

where

$$W_{t_1} \in \mathbb{R}^{|V| \times P}, \qquad W_{t_2} \in \mathbb{R}^{P \times H}. \tag{A.5}$$

For subsequent analysis, define the reconstruction error matrix as

$$\Delta W = W_t - \hat{W}_t. \tag{A.6}$$

For an arbitrary token, let its one-hot representation be $x \in \mathbb{R}^{|V|}$. The token embeddings before and after compression are

$$e = x W_t, \qquad \hat{e} = x \hat{W}_t. \tag{A.7}$$

The analysis focuses on the perturbation introduced by approximating the word embedding matrix. Other input components in BERT, such as positional embeddings and token-type embeddings, are held fixed and therefore do not contribute to the difference term considered below.

A.2 From Matrix Reconstruction Error to Token-Level Representation Error

An important theoretical property of TSVD is that it provides the optimal approximation to the original matrix under a fixed rank constraint. According to the Eckart–Young–Mirsky theorem, $\hat{W}_t$ is the optimal approximation to W_t among all matrices whose rank is at most P:

$$\hat{W}_t = \arg \min_{\mathrm{rank}(B) \leq P} \|W_t - B\|_F. \tag{A.8}$$

Its reconstruction error satisfies

$$\|W_t - \hat{W}_t\|_F^2 = \sum_{i>P} \sigma_i^2, \tag{A.9}$$

and

$$\|W_t - \hat{W}_t\|_2 = \sigma_{P+1}, \tag{A.10}$$

where σ_i denotes the i-th singular value of W_t. These equalities hold when $P < r$; if $P \geq r$, the reconstruction error is zero.

These results show that the component truncated by TSVD is the minimum residual component under the rank-P constraint, rather than an arbitrary deletion of coordinates or embedding dimensions. To further analyze the effect of this matrix-level error on a single token, consider an arbitrary one-hot token input x. The difference between the token embeddings before and after compression is

$$e - \hat{e} = xW_t - x\hat{W}_t = x(W_t - \hat{W}_t) = x\Delta W. \tag{A.11}$$

By the matrix norm inequality,

$$\|e - \hat{e}\|_2 = \|x\Delta W\|_2 \leq \|x\|_2 \|\Delta W\|_2. \tag{A.12}$$

Since the one-hot vector satisfies $\|x\|_2 = 1$, we have

$$\|e - \hat{e}\|_2 \leq \|\Delta W\|_2 = \sigma_{P+1}. \tag{A.13}$$

This inequality indicates that, for any single token, the embedding perturbation introduced by TSVD has an explicit upper bound controlled by the first discarded singular value σ_{P+1}. Therefore, if the singular value spectrum of the word embedding matrix decays clearly after the first P components, the token-level representation error is typically limited and controllable. Intuitively, TSVD compresses the embedding layer by discarding low-energy directions in the spectral tail. If the dominant semantic structure is already concentrated in the leading singular directions, the compressed token representation can still preserve the dominant structure of the original embedding space.

A.3 From Token-Level Representation Error to Downstream Output Error

This section further discusses how the embedding-layer error propagates to downstream encoder outputs. This analysis addresses whether the input-side perturbation could be amplified by subsequent deep layers and adversely affect the final task performance.

For an input sequence of length T, let its one-hot selection matrix be

$$X \in \mathbb{R}^{T \times |V|}. \tag{A.14}$$

The embedding matrices before and after compression are

$$E = XW_t, \qquad \hat{E} = X\hat{W}_t. \tag{A.15}$$

Their difference is

$$E - \hat{E} = X(W_t - \hat{W}_t) = X\Delta W. \tag{A.16}$$

Thus,

$$\|E - \hat{E}\|_F = \|X\Delta W\|_F \leq \|X\|_F \|\Delta W\|_2. \tag{A.17}$$

For a one-hot input matrix X with sequence length T, we have

$$\|X\|_F = \sqrt{T}. \tag{A.18}$$

Therefore,

$$\|E - \hat{E}\|_F \leq \sqrt{T}\,\sigma_{P+1}. \tag{A.19}$$

This shows that the overall perturbation of the embedding layer is jointly controlled by the square root of the sequence length and the magnitude of the spectral-tail singular value, and it does not grow without bound at the input stage.

Furthermore, let the subsequent BERT encoder be abstracted as a mapping function $F(\cdot)$, whose input is the embedding representation E and whose output is the downstream task-related representation $F(E)$. On a bounded local neighborhood of the embedding manifold, the subsequent encoder can be treated as locally Lipschitz continuous. This is a local stability assumption rather than a claim of global non-expansiveness. Under a norm compatible with the Frobenius norm after vectorization, assume that there exists a constant $L > 0$ such that

$$\|F(E) - F(\hat{E})\| \leq L\|E - \hat{E}\|_F. \tag{A.20}$$

Combining the above inequalities gives

$$\|F(E) - F(\hat{E})\| \leq L\sqrt{T}\,\sigma_{P+1}. \tag{A.21}$$

This inequality indicates that, under the stated local Lipschitz assumption, the error introduced by TSVD in the word embedding layer remains controlled when propagated through the subsequent encoder. As long as the chosen truncation rank P preserves the dominant singular values and keeps σ_{P+1} small, the perturbation caused by the low-rank approximation of the word embedding layer can be locally controlled.

It should be noted that the above derivation does not imply that TSVD is lossless for all tasks. If some discriminative information for a specific downstream task is concentrated in low-energy spectral-tail directions, low-rank approximation may still cause performance degradation. Nevertheless, TSVD theoretically guarantees the minimum reconstruction error under a given rank-P constraint. Therefore, its approximation error is the minimum possible error in the sense of optimal low-rank approximation.

A.4 Discussion of Assumptions and Limitations

In summary, the effect of TSVD on the word embedding layer can be understood from two perspectives. First, at the matrix approximation level, TSVD preserves the dominant spectral structure of the original word embedding matrix in an optimal manner, and its error is controlled by the energy of the truncated singular-value tail. Second, at the perturbation propagation level, this matrix-level error can be transformed into bounded token-level representation perturbation and further into bounded downstream output deviation. Therefore, if the word embedding matrix itself exhibits clear spectral decay and the subsequent encoder remains locally stable, applying TSVD to the word embedding matrix usually does not cause excessive performance degradation.

B Empirical Evidence for the Low-Rank and Redundant Structure of the Embedding Matrix

To complement the theoretical analysis in Appendix A, this appendix provides empirical evidence that the embedding matrix exhibits strong redundancy and an approximate low-rank structure,

thereby supporting the use of TSVD in the word embedding layer. The analysis is motivated by two observations. First, many tokens share lexical, semantic, and morphological structures, which can induce strong correlations among their embedding representations. Second, although the word embedding matrix lies in a high-dimensional space, the effective degrees of freedom that carry the dominant embedding variation under the dataset distribution can be smaller than its ambient dimensionality.

Based on these considerations, we analyze the MAVEN dataset and the BERT word embedding matrix from the locally trained model. The embedding matrix used in the experiment is extracted from `bert.embeddings.word_embeddings.weight`, with a shape of 30522×768. In the MAVEN dataset, we collect 4,480 documents, 49,873 sentences, 1,275,644 raw tokens, and 1,434,177 BERT WordPieces.

B.1 WordPiece Sharing and Token Fragmentation in MAVEN

In BERT, raw tokens are decomposed into subword fragments, namely WordPieces, as the basic input units. If different raw tokens share the same WordPiece, this indicates that the vocabulary contains many reusable subword structures. Such shared structures can induce statistical correlations among different rows of the word embedding matrix, thereby providing a basis for low-rank approximation.

Let the raw token set in MAVEN be $\mathcal{T}$ and the BERT WordPiece set be $\mathcal{P}$. For any raw token $t \in \mathcal{T}$, its WordPiece decomposition can be written as

$$t \to \{p_1, p_2, \ldots, p_m\}, \qquad p_i \in \mathcal{P}. \tag{B.1}$$

If multiple raw tokens share the same p_i, then these tokens exhibit structural reuse at the subword level.

In the MAVEN dataset, 52,584 lowercased raw token types are mapped to 23,190 actually occurring WordPiece ids, and each raw token corresponds to 1.124 WordPieces on average. In total, 98,844 token occurrences are split into multiple WordPieces, accounting for 7.75% of all token occurrences. At the type level, 65.85% of raw token types are decomposed into multiple WordPieces. More importantly, 10,785 WordPieces are shared by at least two raw token types, and 41,616 raw token types contain at least one shared WordPiece, accounting for 79.14% of all raw token types.

Figure B.1 shows the token fragmentation distribution of MAVEN under BERT WordPiece. The x-axis denotes the number of WordPieces corresponding to each raw token, and the y-axis denotes the proportion. The blue bars indicate the proportion at the token occurrence level, and the orange bars indicate the proportion at the lowercased token type level.

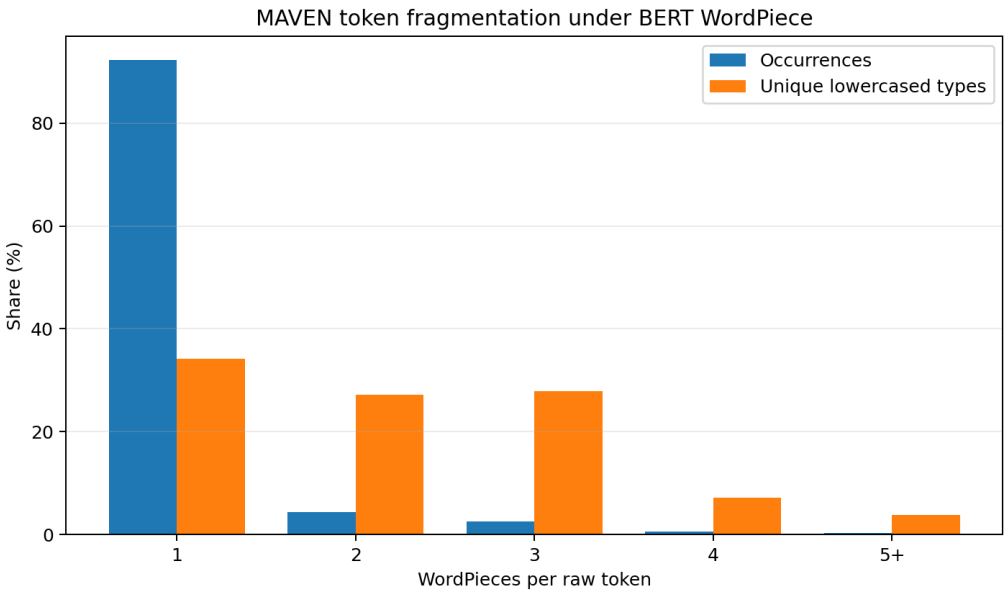

Figure B.1: Token fragmentation in MAVEN under BERT WordPiece.

The above results show that although most token occurrences correspond to a single WordPiece, a large proportion of token types in MAVEN rely on shared subword fragments for representation. This indicates that the word embedding matrix contains substantial shared structures induced by word forms, affixes, and subword reuse.

B.2 Lexical, Morphological, and Semantic Correlations Among Tokens

The redundancy of the word embedding matrix is reflected not only in WordPiece sharing, but also in vector correlations within the embedding space. To verify whether shared structures correspond to higher vector similarity, we construct three token-pair groups: random frequent alphabetic token pairs, token pairs sharing the first five characters, and token pairs sharing the same heuristic stem. In addition, to analyze event-level semantic sharing, we compare the embedding similarities between trigger phrases from different MAVEN event types and trigger phrases from the same MAVEN event type.

For tokens i and j, let their centered embedding vectors be $\tilde{w}_i$ and $\tilde{w}_j$, respectively. Their centered cosine similarity is defined as

$$\cos(\tilde{w}_i, \tilde{w}_j) = \frac{\tilde{w}_i^\top \tilde{w}_j}{\|\tilde{w}_i\|_2 \|\tilde{w}_j\|_2}. \tag{B.2}$$

Here, each raw token or trigger phrase is first mapped to the average of its WordPiece embeddings, then centered by subtracting the frequency-weighted mean computed over the corresponding MAVEN token or trigger-phrase set, and finally normalized with L_2 normalization. The similarity computation and grouping procedure are performed by the analysis script, where the grouping criteria include shared character prefixes, heuristic stems, and MAVEN event types.

The results are shown in Table B.1. The mean centered cosine similarity of random frequent alphabetic token pairs is only 0.0560. In contrast, token pairs sharing the first five characters achieve a mean similarity of 0.3067, approximately 5.48 times that of the random group. Token pairs sharing the same heuristic stem achieve a mean similarity of 0.4971, approximately 8.88 times that of the random group. A similar trend is observed at the event-semantic level: the mean centered cosine similarity of trigger phrases from different event types is 0.0234, whereas trigger phrases from the same MAVEN event type achieve a mean similarity of 0.0894, approximately 3.82 times higher.

Table B.1: Centered cosine similarity under different token-sharing structures.

Token-pair type	Pairs	Mean	Median	P75	P90
Random frequent alphabetic token types	50,000	0.0560	0.0529	0.1179	0.1802
Token types sharing the first five characters	12,329	0.3067	0.2618	0.4998	0.6564
Token types sharing the same heuristic stem	2,852	0.4971	0.5358	0.6348	0.7047
Trigger phrases from different event types	50,000	0.0234	0.0210	0.0765	0.1345
Trigger phrases from the same event type	18,266	0.0894	0.0665	0.1416	0.2399

Figure B.2 further illustrates the same phenomenon. The left panel compares the centered cosine similarity of lexical and morphological token pairs, while the right panel compares the centered cosine similarity of event-semantic trigger phrases. Black dots indicate medians, bars indicate means, and error bars indicate 95% confidence intervals. The results show that token representations in the embedding space are affected by shared lexical, morphological, and event-semantic structures. When token pairs share prefixes, stems, or event types, their centered cosine similarity is substantially higher than that of the random control group.

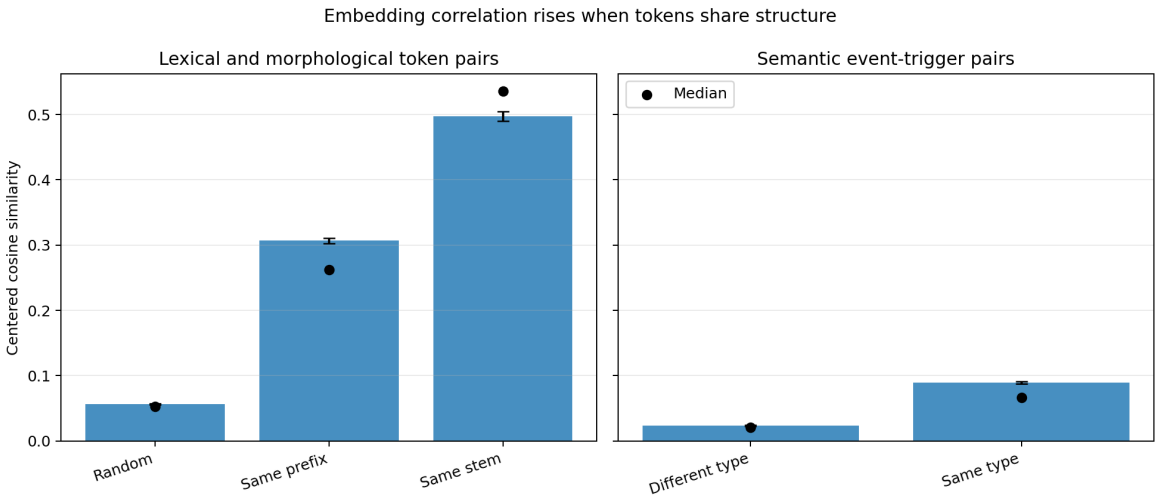

Figure B.2: Increased embedding correlation under shared structures.

These results demonstrate that when tokens share lexical prefixes, morphological stems, or event-semantic categories, their embedding representations exhibit substantially higher correlations. This supports the claim that the word embedding matrix contains clear correlation and redundancy structures, and that token embeddings are distributed around shared lexical, morphological, and semantic patterns. Such correlations provide important empirical evidence for the feasibility of low-rank approximation of the word embedding matrix.

B.3 MAVEN-Frequency Weighted Effective Rank of Embeddings

To further show that the degrees of freedom carrying the dominant dataset-conditioned embedding variation are smaller than the ambient dimensionality, we perform frequency-weighted centering on the WordPiece embeddings actually occurring in MAVEN and analyze their covariance spectrum.

Let w_i denote an actually occurring WordPiece embedding, and let c_i denote its occurrence frequency in MAVEN. The MAVEN-frequency weighted mean is defined as

$$\bar{w} = \frac{\sum_i c_i w_i}{\sum_i c_i}. \tag{B.3}$$

The centered embedding is

$$\tilde{w}_i = w_i - \bar{w}. \tag{B.4}$$

Based on $\tilde{w}_i$, we construct the MAVEN-frequency weighted covariance spectrum and compute the cumulative variance retained under different numbers of principal directions. Let the eigenvalues of the covariance spectrum be

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_H. \tag{B.5}$$

The cumulative centered variance retained at rank P is defined as

$$R(P) = \frac{\sum_{i=1}^{P} \lambda_i}{\sum_{i=1}^{H} \lambda_i}. \tag{B.6}$$

Let $q_i = \lambda_i / \sum_j \lambda_j$ denote the normalized variance distribution. We report the entropy effective rank

$$\text{erank} = \exp\left(-\sum_i q_i \log q_i\right) \tag{B.7}$$

and the participation-ratio rank

$$\text{prank} = \frac{1}{\sum_i q_i^2}. \tag{B.8}$$

These two quantities measure complementary aspects of the effective degrees of freedom in the embedding space. The results show that although the apparent dimensionality of the BERT embedding is 768, the entropy effective rank of the MAVEN-frequency weighted and centered embedding spectrum is 227.8, and the participation-ratio rank is 30.4. Both are substantially lower than the apparent 768-dimensional space. Only 60 dimensions are required to retain 50% of the variance, 237 dimensions are required to retain 75% of the variance, and 445 dimensions are required to retain 90% of the variance, which is still lower than the full 768-dimensional space. The detailed statistics are shown in Table B.2.

Table B.2: Effective-rank statistics of MAVEN-frequency weighted embeddings.

Metric	Value
Apparent embedding dimensionality	768
Actually occurring WordPiece ids	23,190
Total WordPiece occurrences	1,434,177
Entropy effective rank	227.8
Participation-ratio rank	30.4
Dimensions required for 50% variance	60
Dimensions required for 75% variance	237
Dimensions required for 80% variance	294
Dimensions required for 85% variance	362
Dimensions required for 90% variance	445
Dimensions required for 95% variance	554
Dimensions required for 99% variance	691

These results indicate that although the BERT word embedding matrix is embedded in an ambient 768-dimensional space, under the MAVEN data distribution the actually used WordPiece embeddings have effective degrees of freedom that are clearly smaller after frequency weighting and centering. This is reflected by the concentration of the main centered variance in a smaller number of principal directions.

B.4 Reconstruction Quality Under Different Low-Rank Dimensions

To further examine whether low-rank approximation can preserve the embedding structure actually used in MAVEN, we measure the MAVEN-frequency weighted centered variance retained, reconstruction error, and centered reconstruction cosine under different ranks. The results are shown in Table B.3.

Table B.3: MAVEN-frequency weighted centered reconstruction quality under different ranks.

Rank	Variance retained	Reconstruction error	Weighted mean centered cosine
8	29.44%	84.00%	0.5466
16	35.30%	80.43%	0.6085
32	42.30%	75.96%	0.6753
64	50.97%	70.02%	0.7457
96	57.15%	65.46%	0.7890
128	62.19%	61.49%	0.8210
160	66.52%	57.86%	0.8467
192	70.34%	54.46%	0.8679
224	73.75%	51.24%	0.8859
256	76.80%	48.17%	0.9013
320	82.08%	42.33%	0.9266
384	86.49%	36.76%	0.9463
512	93.28%	25.92%	0.9746
768	100.00%	0.00%	1.0000

The results show that as the rank increases, the MAVEN-frequency weighted centered variance retained steadily increases, the reconstruction error steadily decreases, and the centered reconstruction cosine consistently improves. In particular, at rank 256, the model retains 76.80% of the MAVEN-frequency weighted centered variance and achieves a weighted mean centered cosine of 0.9013. At rank 384, the retained variance further increases to 86.49%, and the weighted mean centered cosine reaches 0.9463. At rank 512, the retained variance reaches 93.28%, and the weighted mean centered cosine reaches 0.9746.

Figure B.3 shows the trend of MAVEN-frequency weighted centered variance retained and reconstruction error under different low-rank dimensions. The x-axis denotes the number of embedding components, and the y-axis denotes percentage. The blue curve represents the MAVEN-frequency weighted centered variance retained, while the orange curve represents the MAVEN-frequency weighted centered reconstruction error. The black dashed line indicates the entropy effective rank, and the gray dotted line indicates rank 256. As the rank increases, the variance retained continuously increases, while the reconstruction error continuously decreases. This suggests that the embedding variation actually used under the MAVEN data distribution is concentrated in a relatively smaller number of principal directions. In particular, rank 256 retains 76.80% of the centered variance, and rank 384 retains 86.49%, indicating clear low-rank compressibility of the word embedding matrix.

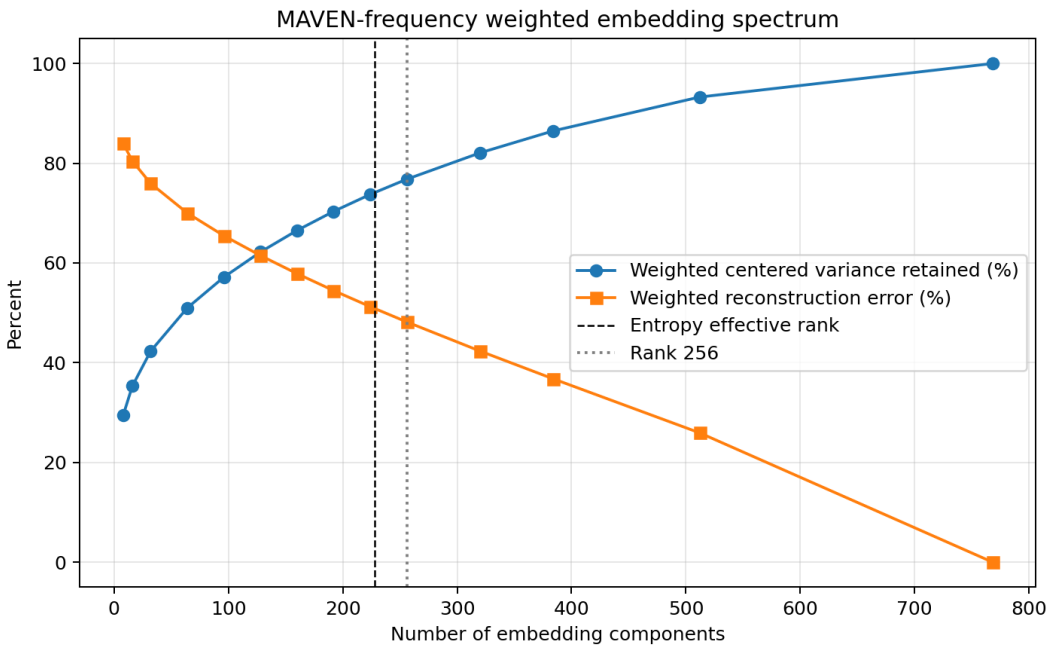

Figure B.3: MAVEN-frequency weighted embedding spectrum and low-rank reconstruction error.

Figure B.4 further examines low-rank reconstruction quality from the perspective of directional consistency. The x-axis denotes the number of low-rank components, and the y-axis denotes the MAVEN-frequency weighted mean centered cosine. The gray dotted line indicates rank 256. The results show that as the rank increases, the MAVEN-frequency weighted mean centered cosine continuously increases. When the rank is 256, this metric reaches 0.9013, indicating that the compressed embeddings preserve high directional consistency with the original embeddings in the dominant directions. This suggests that TSVD can substantially reduce the parameter scale while preserving the main representation structure of the embeddings actually used in MAVEN.

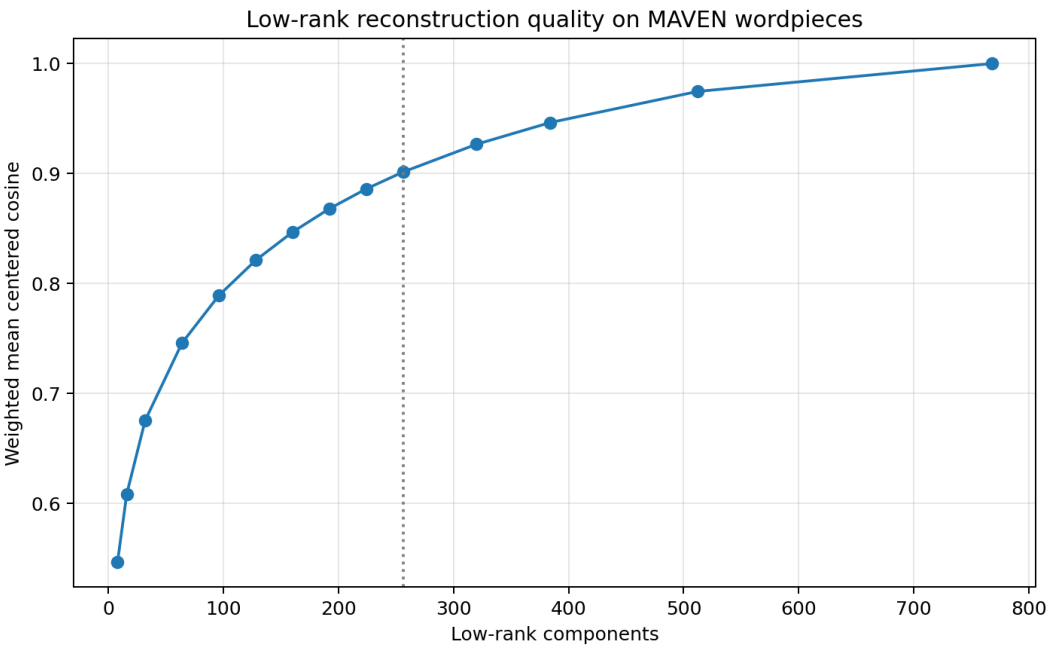

Figure B.4: MAVEN WordPiece reconstruction cosine under different low-rank dimensions.

These results show that, under the MAVEN data distribution, the actually used structure of the BERT word embedding matrix has clear low-rank compressibility. Although the retained variance at relatively low ranks does not fully approach 100%, the directional consistency of reconstruction is already high. As the low-rank dimensionality increases, the centered variance retained increases continuously, the reconstruction error decreases continuously, and the directional consistency between the reconstructed embeddings and the original embeddings improves steadily. In particular, at rank 256, the model retains 76.80% of the MAVEN-frequency weighted centered variance and achieves a weighted mean centered cosine of 0.9013. This indicates that this truncation rank provides a reasonable trade-off between parameter compression and representation preservation.

B.5 Cumulative Variance Analysis of the First 128 Principal Directions

To examine the accumulation of spectral energy in the leading principal directions in more detail, we further plot the cumulative variance curve of the first 128 components. As shown in Figure B.5, the x-axis denotes the component index, and the y-axis denotes the cumulative MAVEN-frequency weighted centered variance. The curve illustrates the cumulative contribution of the leading principal directions to the overall embedding variance. The first 128 principal directions retain 62.19% of the MAVEN-frequency weighted centered variance. Although 128 dimensions are not sufficient to cover most of the variance, they already capture more than half of the centered variation. When the dimensionality is increased to 256, the retained variance increases to 76.80%, and the centered reconstruction cosine increases to 0.9013.

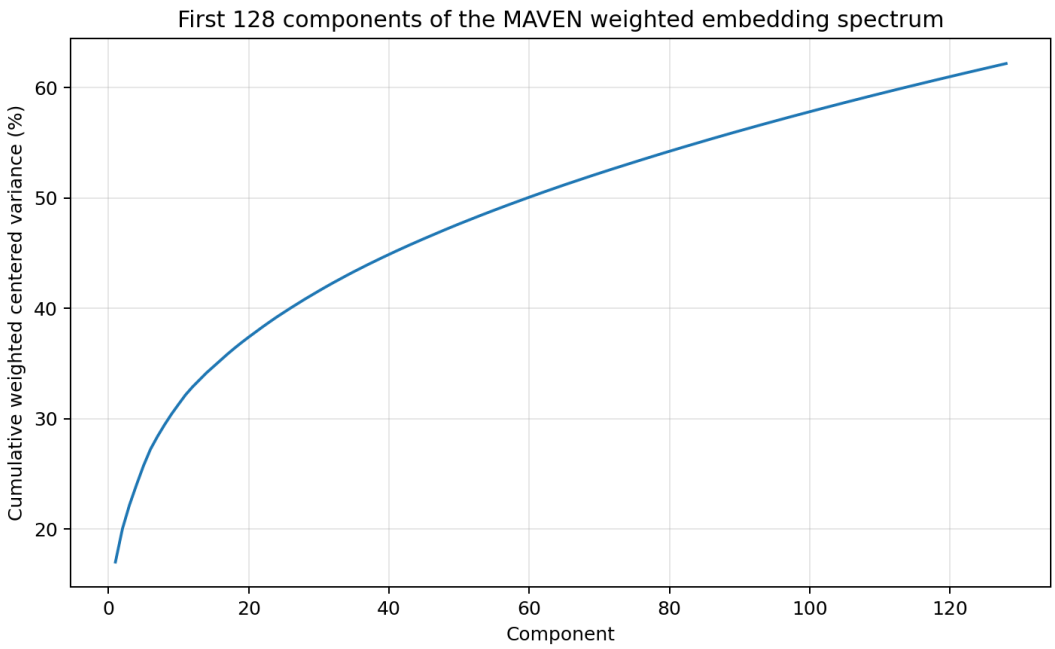

Figure B.5: Cumulative variance curve of the first 128 principal directions.

This result indicates that the MAVEN embedding spectrum exhibits a gradual decay pattern and concentration in the leading principal directions. In other words, the main information of the MAVEN embeddings is preferentially concentrated in the leading directions, providing direct spectral evidence for TSVD-based low-rank compression.

B.6 Summary

This appendix provides an empirical analysis of the BERT word embedding matrix on the MAVEN dataset. The results show that: first, a large number of tokens in MAVEN share WordPiece fragments, and there are clear lexical, morphological, and event-semantic correlations among tokens; second, after MAVEN-frequency weighting and centering, the effective rank of the embeddings is substantially lower than the apparent 768-dimensional space; third, as the number of low-rank components increases, the variance retained and the centered reconstruction cosine improve steadily, with rank 256 already achieving a weighted mean centered cosine of 0.9013. Taken together, these results demonstrate that the word embedding matrix has strong redundancy and an approximate low-rank structure. Combined with Appendix A, the results further indicate that TSVD can remove part of the low-energy and highly redundant tail directions while preserving the dominant spectral directions. As long as the selected truncation rank covers the main spectral energy and maintains high directional consistency in embedding reconstruction, the perturbation introduced by TSVD is generally limited and controllable. This provides empirical support for applying TSVD to compress the word embedding layer.

C Analysis of the TSVD Truncation Rank

C.1 Purpose of the Analysis

In the proposed model, the BERT word embedding matrix is compressed using TSVD, and the truncation rank is set to $P = 256$. This appendix provides an empirical analysis of this setting to justify the choice of the truncation rank. The analysis is based on the `bert-base-uncased` word embedding matrix and the MAVEN dataset used in the experiments.

C.2 Dataset and Tokenization Statistics

This analysis is conducted on the MAVEN corpus after BERT WordPiece tokenization. The overall statistics of the MAVEN corpus used in the analysis are shown in Table C.1.

Table C.1: Overall statistics of the MAVEN corpus used in the truncation-rank analysis.

Item	Value
Documents	4,480
Sentences	49,873
Original tokens	1,275,644
BERT WordPieces	1,434,177
Train + valid events	91,222
Train + valid event mentions	96,897
Event types	168

Under the current sentence-level setting, BERT tokenization is compact. Each raw token is split into approximately 1.124 WordPieces on average, and more than 99.95% of the sentences can be contained within the maximum sequence length of 128. This indicates that the adopted sentence-level input form is consistent with the model configuration used in our experiments. The statistics for each data split are shown in Table C.2.

Table C.2: Split-level statistics of MAVEN after BERT WordPiece tokenization.

Split	Docs	Sentences	Raw tokens	WordPieces	Within max length 128
Train	2,913	32,431	832,186	936,292	99.96%
Valid	710	8,042	204,556	228,819	99.95%
Test	857	9,400	238,902	269,066	99.95%

These statistics show that MAVEN is a large-scale and fine-grained event detection dataset, containing 168 event types and a large number of event trigger mentions. Therefore, when selecting the compression rank of the word embedding matrix, it is not appropriate to use an overly small truncation rank, since doing so may lose lexical and semantic distinctions useful for trigger classification.

C.3 Rank-Sweep Analysis of the BERT Word Embedding Matrix

Let the original BERT word embedding matrix be $W_t \in \mathbb{R}^{|V|\times H}$, where $|V|$ denotes the vocabulary size and H denotes the hidden dimension. TSVD approximates W_t by retaining the first P singular directions:

$$\hat{W}_t(P) = U_P \Sigma_P V_P^\top. \tag{C.1}$$

In the gated TSVD embedding module, this decomposition is further written as

$$W_t \approx AB, \tag{C.2}$$

where

$$A = U_P \Sigma_P \in \mathbb{R}^{|V|\times P}, \qquad B = V_P^\top \in \mathbb{R}^{P\times H}. \tag{C.3}$$

The gated TSVD embedding module introduces token-level gates in the low-rank latent dimension. For token v, the reconstructed embedding can be written as

$$e_v = (A_v \odot g_v)B, \tag{C.4}$$

where $g_v \in \mathbb{R}^P$ is the token-level gate and $\odot$ denotes element-wise multiplication. The parameter count of the original word embedding matrix is

$$|V|H. \tag{C.5}$$

The parameter count of the TSVD-only factorization is

$$|V|P + PH, \tag{C.6}$$

whereas the gated low-rank embedding module has approximately

$$2|V|P + PH \tag{C.7}$$

parameters.

To evaluate the effect of different truncation ranks, we examine the candidate ranks shown in Figure C.1. For each rank, we compute the following metrics: (1) energy retained, namely the proportion of the global spectral energy preserved by the first P singular values; (2) MAVEN weighted cosine similarity, namely the frequency-weighted average cosine similarity between the original embedding and the reconstructed embedding under the MAVEN token distribution; (3) TSVD-only parameter ratio, namely the ratio of the TSVD-only factorization parameters to the original word embedding parameters; and (4) gated parameter ratio, namely the ratio of the gated low-rank embedding module parameters to the original word embedding parameters. These metrics are different from the MAVEN-frequency weighted centered covariance analysis in Appendix B: Appendix B analyzes centered dataset-conditioned embedding variation, whereas this section analyzes TSVD reconstruction and parameter cost of the original BERT embedding matrix.

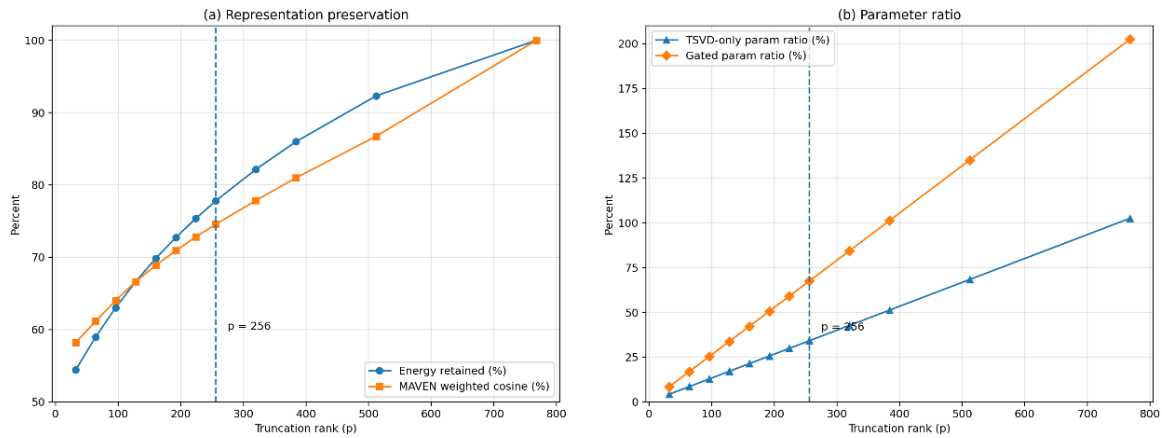

Figure C.1: Rank-sweep analysis of representation preservation and parameter ratio under different TSVD truncation ranks.

As shown in Figure C.1, when $P = 256$, the TSVD approximation retains 77.80% of the global spectral energy of the BERT word embedding matrix. Under the MAVEN token distribution, the frequency-weighted average cosine similarity reaches 0.7456. Compared with more aggressive compression settings, increasing the truncation rank to 256 substantially improves the preservation of MAVEN-related token representations. Although higher truncation ranks can further improve reconstruction quality, they also significantly increase the parameter cost. For example, when the rank is increased from 256 to 384, the parameter ratio of the gated embedding module increases from approximately 67.5% to approximately 101.7%, meaning that the gated low-rank embedding module already exceeds the parameter count of the original embedding table at $P = 384$. Therefore, $P = 256$ provides a more favorable trade-off between reconstruction quality and parameter reduction.

C.4 Representation Preservation for Task-Critical Tokens

Event detection does not rely uniformly on all vocabulary entries. Instead, it depends more heavily on tokens related to event triggers. Therefore, we further evaluate the reconstruction quality on three task-critical token subsets: (1) positive event-trigger WordPieces in the training and validation sets; (2) negative trigger WordPieces in the training and validation sets; and (3) candidate trigger WordPieces in the test set.

When $P = 256$, the results are shown in Table C.3.

Table C.3: Representation preservation for task-critical token subsets at $P = 256$.

Token subset	Total WordPieces	Unique WordPieces	Weighted cosine
Positive event triggers in train and valid	106,067	6,105	0.8288
Negative triggers in train and valid	500,537	20,697	0.8186
Test candidates	139,932	14,736	0.8207

The results show that the truncation rank $P = 256$ preserves the representations of task-critical trigger-related tokens reasonably well. In particular, the weighted cosine similarity of positive event-trigger WordPieces reaches 0.8288. The weighted cosine similarities of negative triggers and test candidate triggers are also higher than 0.81. This is important for MAVEN, because the dataset contains 168 event types and requires fine-grained distinctions among positive triggers, negative triggers, and candidate triggers.

C.5 Summary

In summary, although TSVD compression inevitably introduces some reconstruction error, the rank-256 approximation preserves relatively high representation similarity on the token subsets most relevant to event trigger detection. Therefore, this paper adopts $P = 256$ as an empirical trade-off between embedding compression and representation capacity.